Q1 - Single Correct

If A is involuntary matrix given by $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$, then the inverse of $A/2$ will be

(1) $2A$

(2) $\frac{A^{-1}}{2}$

(3) $\frac{A}{2}$

(4) A^2

Q2 - Single Correct

$\operatorname{tr}(A)$ of a 3×3 matrix $A = (a_{ij})$ is defined by the relation $\operatorname{tr}(A) = a_{11} + a_{22} + a_{33}$ (i.e.$tr(A)$ is sum of the main diagonals elements). Which of the following statement cannot always hold?

(1) $\operatorname{tr}(KA) = K\operatorname{tr}(A)$, ($K$ is scalar)

(2) $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$

(3) $\operatorname{tr}(I_3) = 3$

(4) $\operatorname{tr}(A^2) = [\operatorname{tr}(A)]^2$

Q3 - Single Correct

Let $\Delta_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and Δ_1 denote the determinant formed by the cofactors of elements of Δ_0 and Δ_2 denote the determinant formed by the cofactors at Δ_1 and so on Δ_n denotes the determinant formed by the cofactors at Δ_{n-1}, then the determinant value of Δ_n, denotes the determinant formed by the cofactors at Δ_{n-1}, then the determinant value of Δ_n is

(1) Δ_0^{2n}

(2) $\Delta_0^{2^n}$

(3) $\Delta_0^{n^2}$

(4) Δ^2

Q4 - Single Correct

Matrix A satisfies $A^2 = 2A - I$, where I is the identity matrix, then for $n \geq 2$, A^n is equal to (where, $n \in N$)

(1) $nA - I$

(2) $2^{n-1}A - (n-1)I$

(3) $nA - (n-1)I$

(4) $2^{n-1}A - I$

Q5 - Single Correct

Let three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}, B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then

$\operatorname{tr}(A) + \operatorname{tr}\left(\frac{ABC}{2}\right) + \operatorname{tr}\left(\frac{A(BC)^2}{4}\right) + \operatorname{tr}\left(\frac{A(BC)^3}{8}\right) + \cdots \infty$ is equal to

(1) 6

(2) 9

(3) 12

(4) None of these

Q6 - Single Correct

The determinant $\begin{vmatrix} 1+a+x & a+y & a+z \\ b+x & 1+b+y & b+z \\ c+x & c+y & 1+c+z \end{vmatrix}$ is equal to

(1) $(1+a+b+c)(1+x+y+z) - 3(ax+by+cz)$

(2) $a(x+y) + b(y+z) + c(z+x) - (xy+yz+zx)$

(3) $x(a+b) + y(b+c) + z(c+a) - (ab+bc+ca)$

(4) None of the above

Q7 - Single Correct

In a square matrix A of order 3, the elements, a_{ii}'s are the sum of the roots of the equation

$x^2 - (a+b)x + ab = 0$; $a_{i,i+1}$'s are the product of the roots, $a_{i,i-1}$'s are all unity and the rest of the elements are all zero. The value of the $\det(A)$ is equal to

(1) 0

(2) $(a+b)^3$

(3) a^3-b^3

(4) $\left(a^2+b^2\right)(a+b)$

Q8 - Single Correct

Let D_k be the $k \times k$ matrix with 0's in the main diagonal, unity as the element of Ist row and $(f(k))$ th column and k for all entries. If $f(x)=x-\{x\}$. (where, $\{x\}$ denotes the fractional part function). Then, the value of $\det(D_2)+\det(D_3)$ equals

(1) 32

(2) 34

(3) 36

(4) None of these

Q9 - Single Correct

For a matrix $A=\begin{bmatrix}1 & 2r-1 \\ 0 & 1\end{bmatrix}$, the value of $\prod_{r=1}^{50}\begin{bmatrix}1 & 2r-1 \\ 0 & 1\end{bmatrix}$ is equal to

(1) $\begin{bmatrix}1 & 100 \\ 0 & 1\end{bmatrix}$

(2) $\begin{bmatrix}1 & 4950 \\ 0 & 1\end{bmatrix}$

(3) $\begin{bmatrix}1 & 5050 \\ 0 & 1\end{bmatrix}$

(4) $\begin{bmatrix}1 & 2500 \\ 0 & 1\end{bmatrix}$

Q10 - Single Correct

Let $A=\left\{1^2, 3^2, 5^2, \cdots\right\}$ .9 elements are selected from the set A to make 3×3 matrix, then $\det(A)$ will be divisible by

(1) 9

(2) 36

(3) 8

(4) 64

Q11 - Single Correct

A and B are two square matrices such that $A^2B = BA$ and if $(AB)^{10} = A^k B^{10}$. Then, k is

(1) 1001

(2) 1023

(3) 1042

(4) None of these

Q12 - Single Correct

If α, β, γ are the roots of the equation $x^3 + ax^2 + bx + c = 0$ and if the system of equations

$\alpha u + \beta v + \gamma w = 0, \beta u + \gamma v + \alpha w = 0$ and $\gamma u + \alpha v + \beta w = 0$ posses indeterminable solutions, then

$a^3 + 6c$ equals

(1) ab

(2) $2ab$

(3) $3ab$

(4) None of these

Q13 - Single Correct

If $\begin{vmatrix} a & b & 1 \\ b & c & 1 \\ c & a & 1 \end{vmatrix} = 2010$ and $\begin{vmatrix} c-a & c-b & ab \\ a-b & a-c & bc \\ b-c & b-a & ca \end{vmatrix} - \begin{vmatrix} c-a & c-b & c^2 \\ a-b & a-c & a^2 \\ b-c & b-a & b^2 \end{vmatrix} = p$, then the number of positive

divisors of the number P is

(1) 36

(2) 49

(3) 64

(4) 81

Q14 - Single Correct

Let A, B, are square matrices of same order satisfying $AB = A$ and $BA = B$, then $\left(A^{2010} + B^{2010}\right)^{2011}$ equals

(1) $A + B$

(2) $2010(A + B)$

(3) $2011(A + B)$

(4) $2^{2010}(A + B)$

Q15 - Single Correct

Let A be a 3×3 matrix given by $A = \left(a_{ij}\right)_{3\times3}$. If for every column vector X satisfies $X^T AX = 0$ and $a_{12} = 2008, a_{13} = 2010$ and $a_{23} = -2012$. Then, the value of $a_{21} + a_{31} + a_{32}$ is

(1) -6

(2) 2006

(3) -2006

(4) 0

Q16 - Multiple Correct

Let A and B are two square idempotent matrices such that $AB \pm BA$ is a null matrix, then the value of the det $|A - B|$ can be equal to

(1) -1

(2) 1

(3) 0

(4) 2

Q17 - Multiple Correct

A square matrix A with elements from the set of real numbers is said to be orthogonal, if $A' = A^{-1}$. If A is an orthogonal matrix, then

(1) A^T is orthogonal

(2) A^{-1} is orthogonal

(3) $\operatorname{adj}(A) = A^T$

(4) $|A^{-1}| = 1$

Q18 - Multiple Correct

D is a 3×3 diagonal matrix. Which of the following statement is not true?

(1) $D^T = D$

(2) $AD = DA$ for every matrix A of order 3×3

(3) D^{-1} if exists is a scalar matrix

(4) None of the above

Q19 - Multiple Correct

If A and B are two 3×3 matrices such that their product AB is a null matrix, then

(1) $\det A \neq 0 \Rightarrow B$ must be a null matrix

(2) $\det B \neq 0 \Rightarrow A$ must be a null matrix

(3) If none of A and B are null matrices, then atleast one of the two matrices must be singular

(4) If neither $\det A$ nor $\det B$ is zero, then the given statement is not possible

Q20 - Multiple Correct

If there are three square matrices A, B, C of same order satisfying the equation $A^2 = A^{-1}$ and $B = A^{2^n}$ and $C = A^{2^{(n-2)}}$, then which of the following statements are true?

(1) $|B - C| = 0$

(2) $(B + C)(B - C) = 0$

(3) B must be equal to C

(4) None of these

Q21 - Multiple Correct

If p, q, r, s are in an AP and $f(x) = \begin{vmatrix} p+\sin x & q+\sin x & p-r+\sin x \\ q+\sin x & r+\sin x & -1+\sin x \\ r+\sin x & s+\sin x & s-q+\sin x \end{vmatrix}$ such that $\int_0^2 f(x)dx = -4$, then the common difference of then AP can be

(1) -1

(2) $1/2$

(3) 1

(4) 2

Q22 - Multiple Correct

If a system of the equation $(\alpha+1)^3 x + (\alpha+2)^3 y - (\alpha+3)^3 = 0$ and

$(\alpha+1)x + (\alpha+2)y - (\alpha+3) = 0, x + y - 1 = 0$ is consistent, what is the value of α?

(1) 1

(2) 0

(3) -3

(4) -2

Q23 - Multiple Correct

If $\Delta = \begin{vmatrix} x & 2y-z & -z \\ y & 2x-z & -z \\ z & 2y-z & 2x-2y-z \end{vmatrix}$, then

(1) $(x-y)$ is a factor of Δ

(2) $(x-y)^2$ is a factor of Δ

(3) $(x-y)^3$ is a factor of Δ

(4) Δ is dependent on Z

Q24 - Multiple Correct

Let $a, b > 0$ and $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then

(1) $(a+b-x)$ is a factor of Δ

(2) $x^2 + (a+b)x + a^2 + b^2 - ab$ is a factor of Δ

(3) $\Delta = 0$ has three real roots, if $a = b$

(4) None of the above

Q25 - Multiple Correct

If $\Delta = \begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ b\alpha + c & c\alpha + d & a\alpha^3 - c\alpha \end{vmatrix} = 0$, if

(1) b, c, d are in an AP

(2) b, c, d are in GP

(3) b, c, d are in HP

(4) α is root of $ax^3 - bx^2 - 3cx - d = 0$

Q26 - Multiple Correct

Let $\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$, then

(1) $(a+b)$ is a factor of Δ

(2) $(a+2b)$ is a factor of Δ

(3) $(2a+3b)$ is a factor of Δ

(4) a^2 is a factor of Δ

Q27 - Multiple Correct

If for the matrix $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$; $|A^3| = 125$, then the value of α is

(1) 1

(2) -1

(3) 3

(4) -3

Q28 - Paragraph 1

Passage I (For Question 28, 29)

If A is a symmetric and B is a skew-symmetric matrix and $A+B$ is non-singular and $C=(A+B)^{-1}(A-B)$, then

$C^T(A+B)C$, is

(1) $A+B$

(2) $A-B$

(3) A

(4) B

Q29 - Paragraph 1

$C^T(A-B)C$, is

(1) $A+B$

(2) $A-B$

(3) A

(4) B

Q30 - Paragraph 2

Passage II (For Question 30, 31)

3×3 matrices are formed using the elements from the set $\{-3,-2,-1,0,1,2,3\}$, then

Number of matrices that are skew-symmetric matrices (or) scalar matrices is

(1) $7^7 - 7^4$

(2) $7^7 - 7^3$

(3) $7^3 + 6$

(4) $7^4\left(7^3 - 6\right)$

Q31 - Paragraph 2

Number of matrices that are lower diagonal (or) upper diagonal is

(1) $7^6 - 7^3$

(2) $7^6 - 2\left(7^3\right)$

(3) $7^3\left(2 \times 7^3 - 1\right)$

(4) $7^9 - 7^6$

Q32 - Paragraph 3

Passage III (For Question 32, 33)

Let A be a square matrix of order 2 or 3 and l be the identity matrix of the same order, then the matrix $A - \lambda l$ is called characteristic matrix of the matrix A, where λ is some complex number. The determinant of the characteristic matrix is called characteristic determinant of the matrix A which will of course be a polynomial of degree 3 in λ.

The equation $|A - \lambda l| = 0$ is called the characteristic equation of the matrix A and its roots are called characteristic roots or eigenvalues. It is known that every square matrix satisfies the characteristic equation.

If one of the eigenvalues of a square matrix A of order 3×3 is zero, then

(1) det A must be non-zero

(2) det A must be zero

(3) adj A must be a zero matrix

(4) None of the above

Q33 - Paragraph 3

If A' denotes transpose of matrix A, $A'A = I$ and $\det A = 1$, then $|\det(A - I)|$ must be equal to

(1) 0

(2) -1

(3) 1

(4) None of these

Q34 - Paragraph 4

Passage IV (For Question 34, 35)

If A is a square matrix, then the equation in λ obtained by $|A - \lambda| = 0$ is called as characteristic equation.

Every matrix satisfies its own characteristic equation. Given that $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$. Then,

If $A^3 + kA^2 + lA + mI = 0$. Then, $(k + m + l)$ is

(1) 0

(2) 9

(3) 5

(4) 6

Q35 - Paragraph 4

Inverse of A is

(1) $A^2 + 3A + I$

(2) $A^2 - 3A - I$

(3) $-1/9\left(A^2 - 3A - I\right)$

(4) None of these

Q36 - Paragraph 5

Passage V (For Question 36, 37)

Let A be the set of all 3×3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0.

The number of matrices in A, is

(1) 12

(2) 6

(3) 9

(4) 3

Q37 - Paragraph 5

The number of matrices of A for which the system of linear equations $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, is

(1) less than 4

(2) atleast 4 but less than 7

(3) atleast 7 but less than 10

(4) atleast 10

Q38 - Integer Type

If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\det(A^n - I) = 1 - \lambda^n, n \in N$. Then, the value of λ, is

Q39 - Integer Type

A square matrix P satisfies $P^2 = I - P$, where I is an identity matrix. If $P^n = 5I - 8P$, then n is

Q40 - Integer Type

Let A be an idempotent matrix satisfying $(I-0.4A)^{-1}=I-\lambda A$, where I is unit matrix of the same order as A. Then, the value of $|3\lambda|$ is

Q41 - Integer Type

The number of 2×2 matrices A, that are there with the elements as real numbers satisfying $A+A^{T}=I$ and $AA^{T}=I$ is

Q42 - Integer Type

The number of orthogonal matrices of order 2×2 whose entries are 0,1 is

Q43 - Integer Type

Matrix A is given by $A=\begin{bmatrix}3 & 11\\ 2 & 8\end{bmatrix}$, then the determinant of $A^{2011}-5A^{2010}$ is $|\lambda\cdot 2^{2012}|$, then λ is

Q44 - Integer Type

Suppose A is a matrix such that $A^2=A$ and $(I+A)^{10}=I+kA$, where $k=2\cdot 2^{\lambda}-1$. Then, the value of λ is

Q45 - Integer Type

For any 2×2 matrix A, if $A(\operatorname{adj} A)=\begin{bmatrix}10 & 0\\ 0 & 10\end{bmatrix}$, then the value of $|A|$ is

Q46 - Integer Type

Let k be a positive real number and

$$A=\begin{bmatrix}2k-1 & 2\sqrt{k} & 2\sqrt{k}\\ 2\sqrt{k} & 1 & -2k\\ -2\sqrt{k} & 2k & -1\end{bmatrix}$$

$$B=\begin{bmatrix}0 & 2k-1 & \sqrt{k}\\ 1-2k & 0 & 2\sqrt{k}\\ -\sqrt{k} & -2\sqrt{k} & 0\end{bmatrix}$$

If $\det(\operatorname{adj} A)+\det(\operatorname{adj} B)=10^6$, then the value of $[k]$ is

Q47 - Integer Type

Let the matrix A and B be defined as $A=\begin{bmatrix}3 & 2\\ 2 & 1\end{bmatrix}$ and $B=\begin{bmatrix}3 & 1\\ 7 & 3\end{bmatrix}$. Then, the value of $\left|\det\left(2A^9B^{-1}\right)\right|$ is

Q48 - Integer Type

Let $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & 4\sin 2x\\ \sin^2 x & 1+\cos^2 x & 4\sin 2x\\ \sin^2 x & \cos^2 x & 1+4\sin 2x\end{vmatrix}$, then the maximum value of $f(x)$, is

Q49 - Integer Type

If A, B and C are $n\times n$ matrices and $\det(A)=2, \det(B)=3$ and $\det(C)=5$, then the value of $\frac{5}{3}\det\left(A^2BC^{-1}\right)$ is equal to

Answer Key

Q1 (1)	**Q2** (4)	**Q3** (2)	**Q4** (3)
Q5 (1)	**Q6** (1)	**Q7** (4)	**Q8** (2)
Q9 (4)	**Q10** (3)	**Q11** (2)	**Q12** (4)
Q13 (4)	**Q14** (4)	**Q15** (3)	**Q16** (1, 2, 3)
Q17 (1, 2)	**Q18** (4)	**Q19** (1, 2, 3, 4)	**Q20** (1, 2, 3)
Q21 (1, 3)	**Q22** (4)	**Q23** (1, 4)	**Q24** (1, 2, 3)
Q25 (2, 4)	**Q26** (1, 2)	**Q27** (3, 4)	**Q28** (1)
Q29 (2)	**Q30** (3)	**Q31** (3)	**Q32** (2)
Q33 (4)	**Q34** (3)	**Q35** (3)	**Q36** (1)
Q37 (2)	**Q38** (2)	**Q39** (6)	**Q40** (2)
Q41 (2)	**Q42** (2)	**Q43** (7)	**Q44** (9)
Q45 (10)	**Q46** (4)	**Q47** (2)	**Q48** (6)
Q49 (4)			